

Short proof of the sharp first derivative case of van der Corput's lemma

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Lemma 1 (Second mean value theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be nonnegative and nonincreasing and $g : [a, b] \rightarrow \mathbb{R}$ integrable. Then there exists $\xi \in [a, b]$ such that*

$$\int_a^b f(x)g(x) dx = f(a) \int_a^\xi g(x) dx.$$

Lemma 2. *If $f : [a, b] \rightarrow \mathbb{R}$ with $f \geq 0$ and f nonincreasing, and $g : [a, b] \rightarrow \mathbb{C}$, then*

$$\left| \int_a^b f(x)g(x) dx \right| \leq f(a) \sup_{\xi \in [a, b]} \left| \int_a^\xi g(x) dx \right|$$

Remark. This continues to hold for Banach-space valued g (by the Hahn–Banach theorem).

Proof. This follows from the second mean value theorem for integrals, Lemma 1. □

The following sharp version of the first derivative case of van der Corput's lemma is proved in [1, Cor. 5] using a complex version of the second mean value theorem, which gives a short and elegant proof. The proof can be further shortened by instead using a mean-value inequality, Lemma 2.

Lemma 3 (van der Corput's lemma, order one case). *Let $\phi : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Assume that ϕ' is increasing with $\phi' \geq \lambda > 0$ on (a, b) . Then*

$$\left| \int_a^b e^{i\phi(t)} dt \right| \leq \frac{2}{\lambda}.$$

Remark. The constant 2 is sharp in this inequality (consider $\phi(t) = t$ on $(a, b) = (0, \pi)$).

Proof. Changing variables,

$$\int_a^b e^{i\phi(t)} dt = \int_{\phi(a)}^{\phi(b)} e^{iy} \frac{1}{\phi'(\phi^{-1}(y))} dy = \int_{\phi(a)}^{\phi(b)} e^{iy} g(y) dy,$$

where $g(y) = 1/(\phi'(\phi^{-1}(y)))$ for $y \in (\phi(a), \phi(b))$ and $g(\phi(a)) = 1/\lambda$, $g(\phi(b)) = 0$. Since g is positive and nonincreasing on $[a, b]$, we may apply Lemma 2 and obtain

$$\left| \int_a^b e^{i\phi(t)} dt \right| \leq \frac{1}{\lambda} \sup_{\xi \in [a, b]} \left| \int_a^\xi e^{iy} dy \right| \leq \frac{2}{\lambda},$$

since $\left| \int_a^\xi e^{iy} dy \right| = |e^{ia} - e^{i\xi}| \leq 2$ and $\phi'(a) > \lambda$. □

References

- [1] K. Rogers. Sharp van der Corput estimates and minimal divided differences. arXiv:0311013, 2003.